

Test Normal Cases

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Build a Paper Prototype

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose Inputs and Outputs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important Outputs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Run a fair comparison for Testing routine cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Testing routine cases. Proposed explanation: “A correct label is always a full pass”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit a new test record

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10. Select the first item with capacity at least the group size, or no match; invalid input needs clarification. The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”. Write all expected results, mark full passes, calculate the fraction and explain the failure. Give one untested input category. No booking occurs.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

6. Solve the given case

Four routine cases are chosen before testing. Three are correct.

What can be reported?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Test a sorter on two clear pencil cards and two clear book cards. It gets one book wrong. Report the result and keep the failed case.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 33 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create and run the sorter's normal-case test plan with unique IDs and version label.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Prepare the four cases

Create the test record for R1 11 Yes, R2 9 No, R3 6 No and R4 9 Yes. Before running anything, write each expected output, explanation and purpose using catalogue B version 1. Include blank places for observed output, observed reason, pass or fail and trace reference. State the version labels and pass criterion.

My reasoning and evidence

13. W2 Run a fair comparison

Operate your saved P1 cards and the manual baseline on all four R cases. Record actual results and traces on your sheet. Summarise full passes over all attempted cases for each method, name any failures and explain why a tie would not prove the prototype improves correctness. Do not invent an observed run.

My reasoning and evidence

14. W3 Keep the failed case in the total

Use the fictional Q version 0 record. Calculate its full-pass fraction and the baseline's fraction. Explain why R4 fails and why dropping it to report 3 of 3 misrepresents the original four-case set. State what evidence you would inspect before changing Q.

My reasoning and evidence

15. W4 Audit a new test record

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10. Select the first item with capacity at least the group size, or no match; invalid input needs clarification. The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”. Write all expected results, mark full passes, calculate the fraction and explain the failure. Give one untested input category. No booking occurs.

My reasoning and evidence

16. W5 State what the evidence leaves open

Explain why catalogue B cannot produce B3 or B4 as its first fitting output even if you try many ordinary requests. Distinguish checking a row during a failed search from successfully selecting it. Write a limited conclusion for a P1 run that passed all four R cases and name a useful next test category.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Test Normal Cases

A fictional pencil-pouch finder has been run on four ordinary requests.

Catalogue: P1 capacity 3, P2 capacity 7; first fitting pouch. Valid whole counts 1–9.

Requests 2,4,6,8; recorded outputs P1,P2,P1,none. These are invented records, not class results.

AI ACADEMY / AI-1/33

Expected versus recorded routine outputs

	Request	Expected	Recorded	Result
Test 1	2	P1	P1	pass
Test 2	4	P2	P2	pass
Test 3	6	P2	P1	fail
Test 4	8	none	none	pass

Read a cell using both its row and column labels.

Test Normal Cases

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write expectations

For 2,4,6,8 expect P1,P2,P2,none.



2. Compare outputs

Runs at 2,4 and 8 agree; run at 6 disagrees.



3. Trace failure

P1 holds only 3, so it cannot satisfy request 6.



4. Report

Three of four routine cases pass; retain the failure.

CHANGE ONE THING

Prototype reports none for request 6 instead of P1.

Explain the changed result using the matching lesson.

Why is none correct for 8 but incorrect for 6? Use both capacity cards.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Maintain sorter tests with input, independently derived expectation, actual output, version and full-pass judgment.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Maintain sorter tests with input, independently derived expectation, actual output, version and full-pass judgment.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional pencil-pouch finder has been run on four ordinary requests.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three of four routine cases pass; retain the failure.

1. Trace failure
2. Write expectations
3. Compare outputs
4. Report

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report Three of four routine cases pass; retain the failure. An ordinary case is not automatically correct. Define expected outputs from the contract before reading prototype outputs. Checking both selected ID and eligibility catches the wrong P1 suggestion. A no-match output can be a full pass when that is exactly what the valid request requires.

Practice 2: explain the deciding evidence

Changed-case practice: Prototype reports none for request 6 instead of P1. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. That run still fails.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That run still fails. P2 can hold 6, so none is also wrong under the first-fit rule.

Repair a missing building block

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the repair?. Return to this week and explain what you changed.

Review Choose Inputs and Outputs (ai-1/30). Goal: Turn the project outcome into precise input and output requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?. Return to this week and explain what you changed.

Review Verify Important Outputs (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant evidence. Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
